PROJECT REPORT

Movie Recommendation System Using Content-Based Filtering

Submitted by:

AI/ML Team

Team Members:
Kanishk Upadhayay
Rohit Kumar
Aaliya Nahid
Krishna Dubey
Astha Sinha

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CERTIFICATE

This is to certify that the project report titled “Movie Recommendation System Using Content-Based Filtering” submitted by the AI/ML Team, in partial fulfilment of the requirements for internship/project evaluation, is a bonafide work carried out under my supervision.

Project Guide
Name – Mr. Aditya Bangia
Designation – AI Faculty
Department – AI/ML

Date: 27 – 11 – 2025

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our guide Mr. Aditya Bangia for providing constant guidance and support throughout the project. We also thank our institution and team members for their contributions.

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1. INTRODUCTION

A movie recommendation system helps users discover relevant films based on their preferences. Content-based filtering analyzes movie attributes such as genres, keywords, cast, and descriptions to generate personalized suggestions.

2. PROBLEM STATEMENT

With thousands of movies available, users need a system that identifies films matching their interests. The challenge is to design a model that accurately recommends movies similar to those the user already likes.

3. OBJECTIVES

- Build a content-based movie recommendation system.
- Extract and vectorize movie features.
- Evaluate the performance of the recommendation model.
- Provide accurate and relevant movie suggestions.

4. LITERATURE REVIEW

Content-based filtering relies on analyzing item features instead of user-user interaction. TF-IDF vectorization and cosine similarity are widely used for movie recommendation models.

5. SYSTEM ARCHITECTURE

1. Data Layer – Movie metadata (genres, overview, cast, crew).
2. Feature Engineering – TF-IDF and one■hot encoding.
3. Similarity Engine – Cosine similarity matrix.
4. Application Layer – User input and movie output.

6. METHODOLOGY

- Feature extraction using TF-IDF.
- Cosine similarity computation.
- Ranking the most similar movies.
- Displaying the recommended results.

7. DATA COLLECTION & PREPARATION

Dataset sourced from IMDb/MovieLens.
Steps:

• Cleaning missing values
• Encoding genres
• Vector creation using TF-IDF

8. IMPLEMENTATION & CODE

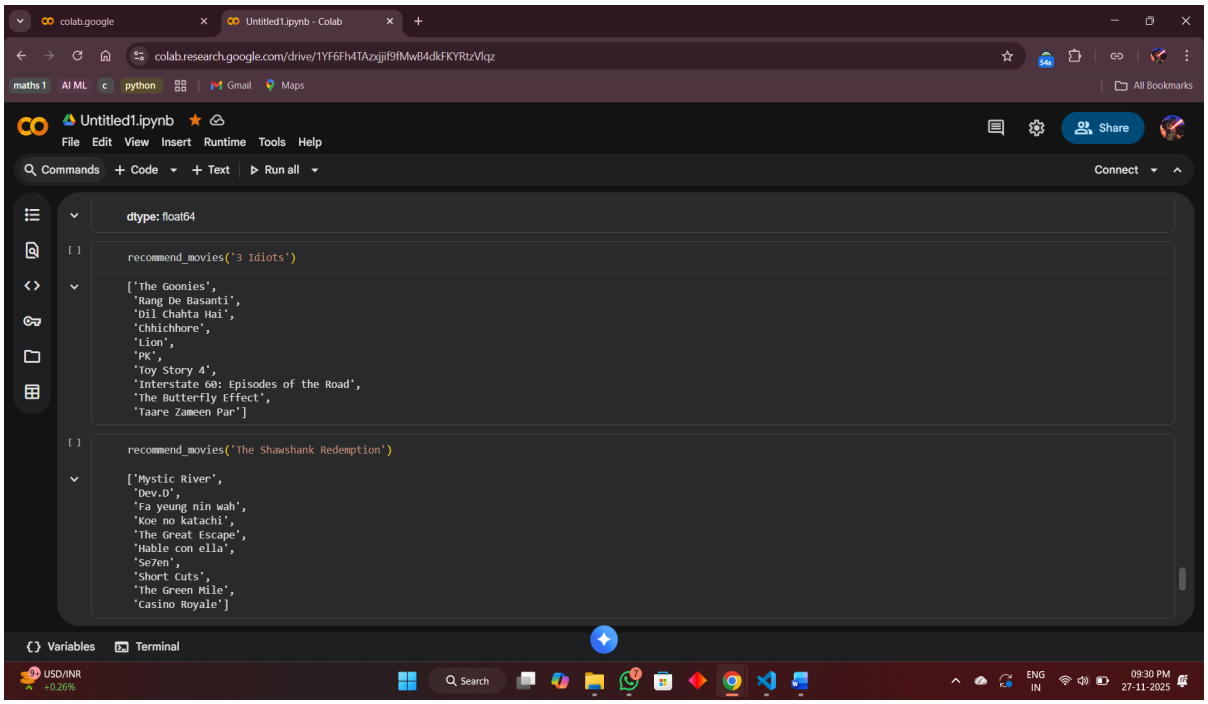
• TF-IDF Vectorizer – Extracts important words from movie descriptions.
• Cosine Similarity – Computes how similar movies are.
• Pandas, NumPy, Scikit■learn used for model development.

9. EVALUATION METRICS

• Precision@K
• Recall@K
• Coverage
• Diversity

10. OUTPUT SCREENSHOTS

Below is the actual screenshot of the system running in Google Colab.



11. RESULTS & DISCUSSION

Content-based filtering performs well but may lack diversity because it relies only on item similarity.

12. CONCLUSION

The system successfully recommends movies using content-based filtering. Future enhancements may combine collaborative filtering for improved accuracy.

13. REFERENCES

• Scikit-learn Documentation
• IMDb Dataset
• Research papers on recommendation systems